



Research Paper

Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger

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ABSTRACT

The purpose of this study is to investigate the freezing phenomenon in Printed Circuit Heat Exchanger (PCHE) for cryogenic applications. A high heat flux in PCHE can lower the temperature of the hot fluid below the freezing point when heat is transferred from the cold fluid, thus initiating the freezing phenomenon. Hence, a thorough study is required to understand the freezing phenomenon for the use of PCHE in cryogenic conditions without degradation. For this, experiments on the hydraulic and thermal performance of PCHE were conducted to figure out the conditions that lead to the freezing phenomenon. Four types of lab-scale PCHEs with semi-circular channels and diameter range 1 to 3 mm, were fabricated using STS316L. Nitrogen at -166 to -140°C and ethylene-glycol water (50 wt%) at 40°C were used as a working fluid in the hot and cold channels. Reynolds number ranged from 4082 to 11,794 for the cold channel and 0 to 441 for the hot channel. Based on the experimental data, Freeze Safety Region (FSR) was proposed as a criterion to compare the risk level of freezing phenomenon occurring within PCHEs. When the Reynolds number of the hot channel was 100, FSR increased by up to 7.3 times as the flow rate of the cold channel was reduced by 50 %. In addition, FSR increased by up to 3.3 times as the diameter of the hot channel was enlarged by 1.5 times. These provide valuable insights into the prevention of freezing phenomenon in PCHE.

1. Introduction

Printed Circuit Heat Exchanger (PCHE) has recently received much attention across various industrial sectors, mainly due to their compatibility with extreme conditions, including cryogenic, ultra-high temperature, and ultra-high pressure applications. PCHE is a compact heat exchanger with an exceptionally high heat transfer area per unit volume [1] in which the base substrate is etched to create mini/micro-sized channels, which are then strongly bonded through diffusion bonding across a number of layers. Heat transfer through the internal mini/microchannels achieves high heat transfer area, resulting in a high heat transfer efficiency. Consequently, PCHE is typically regarded as having the potential to reduce the size by a factor of one-fifth to one-sixth in compared to shell and tube-type heat exchanger [2]. Furthermore, PCHE exhibit a high degree of structural freedom and is constructed from a

single material, demonstrating excellent structural stability.

After its invention in Australia in the 1980s [3], many studies have been conducted experimentally and numerically to figure out the thermal and hydraulic performance of PCHEs in various application. In particular, PCHE have been used in supercritical CO_2 cycle loop and high temperature reactors. Nikitin et al. [4] and Tsuzuki et al. [5] conducted experimental and numerical analyses of the flow and thermal performance of PCHEs used in the CO_2 cycle loop. In addition, Baik et al. [6] and Kim et al. [7] proposed Nusselt number correlations for supercritical CO_2 by using numerical analyses. Kim et al. [8] developed unified correlations of fanning factor and Nusselt number through the experiment and the numerical analysis of the precooler in high temperature reactor. Li et al. [9] used machine learning methods to predict the heat transfer coefficient and pressure drop along the straight channel of PCHE during supercritical methane flow. As the performance of the PCHE is influenced by the type of channel, various channel designs have

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Nomenclature			
A	area [m ²]	Ste	Stefan number [-]
A_{cross}	cross sectional area [m ²]	t	time [s]
A_{HT}	heat transfer area [m ²]	T	temperature [°C]
Bi	Biot number [-]	T_f	freezing point [°C]
c_n	heat capacity ratio [-]	ΔT_{LMTD}	log mean temperature difference [K]
C	heat capacity rate [W/K]	u	flow velocity [m/s]
C_p	specific heat [kJ/kg • K]	U	overall heat transfer coefficient [W/m ² • K]
D	diameter [m]	z	axial co-ordinate [-]
EGW	ethylene-glycol water [-]	Z	non-dimensional axial coordinate [-]
f	friction factor [-]	Greek Symbols	
h	convective heat transfer coefficient [W/m ² • K]	α	thermal diffusivity [m ² /s]
h_{ls}	heat of liquefaction [W]	δ	thickness [m]
i	enthalpy [J/kg]	ϵ	effectiveness [-]
k	thermal conductivity [W/m • K]	θ	non-dimensional temperature [-]
l	non-dimensional length [-]	μ	dynamic viscosity [Pa • s]
L	length [m]	ν	kinematic viscosity [m ² /s]
$L_{channel}$	length of the channel [m]	ρ	density [kg/m ³]
$\dot{m}$	mass flow rate [kg/min]	σ	pitch [m]
N	number [-]	τ	non-dimensional time [-]
N_2	nitrogen [-]	Subscripts	
NTU	number of transfer units [-]	cold	cold channel
Nu	Nusselt number [-]	f	location of freeze-front
P	pressure [kPa]	freeze	freezing standard
PGW	propylene-glycol water [-]	FSR	freeze safety region
ΔP	differential pressure [kPa]	hot	hot channel
Pe	Peclet number [-]	in	inlet
PP	pumping power	l	liquid
Pr	Prandtl number [-]	max	maximum value
Q	heat transfer rate [W]	n	n th node
r	radial co-ordinate [-]	out	outlet
R	non-dimensional radial co-ordinate [-]	tot	total
Re	Reynolds number [-]	w	wall

been investigated. These include straight channels [10–12], zigzag channels [13,14], S-shaped channels [15], and airfoil channels [16], all of which have been proposed to enhance the performance of PCHE.

Since PCHEs have been used in cryogenic environments such as LNG processes [17] and hydrogen refueling stations [18], there are difficulties when developing PCHEs in cryogenic condition compare to PCHEs in normal condition. In cryogenic environments, PCHEs are more sensitive to axial conduction, changes in physical properties and heat loss than in normal environments [19,20]. Due to these difficulties, in the case of PCHEs for cryogenic application, a relatively small number of studies have been conducted compared to the case of PCHEs for CO₂ cycle loop and high temperature reactor. Kwon et al. [21] conducted experimental studies to determine the thermal performance of PCHE at cryogenic temperatures. They investigated single-phase and two-phase flow as well as the heat transfer characteristics of nitrogen at low-temperature inside a PCHE and proposed empirical correlations for Nusselt number. Shin et al. [22,23] used supercritical nitrogen and gaseous nitrogen as working fluids in PCHE to identify the heat transfer characteristics of supercritical fluids and suggested the empirical correlation for Nusselt number. Moreover, Shin et al. [24] conducted experiments on a large-scale PCHE in the supercritical low temperature region and reported modified correlation for Nusselt number. Zhao et al. [25] conducted experiments on heat transfer between supercritical nitrogen and R22 inside a PCHE. In their study, a correlation for the heat transfer coefficient of supercritical nitrogen was proposed using CFD and compared with experimental data.

Most studies have focused on the performance of PCHEs, such as the

heat transfer coefficient and efficiency. However, when the objective of a PCHE is to vaporize a cryogenic fluid, the freezing phenomenon of the hot fluid can be detrimental to the operation of the heat exchanger. In liquid hydrogen applications, the temperature of cryogenic fluids is expected to be lower than in LNG applications, leading to a high probability of freezing phenomenon of the hot fluid. In our previous research [26], freezing experiments were conducted using a lab-scale PCHE. The pressure drop in the hot channel was measured to detect the solid layer, and allowable mass flow rates were proposed to avoid the freezing phenomenon. While our previous research provides insight to avoid freezing phenomenon, there are some limitations. The freezing experiments were conducted with a single PCHE, making it difficult to explain the effect of channel geometries. Additionally, a dimensionless parameter which represents the freezing phenomenon inside PCHEs is required to provide a more comprehensive guideline for avoiding freezing phenomenon.

In this study, experiments were conducted to analyze the conditions under which the freezing phenomenon of the hot fluid occurs in PCHEs. To achieve this four types of lab-scale PCHEs were fabricated, and experiments were performed using low-temperature gaseous nitrogen (N_2) and Ethylene–Glycol Water (EGW) as the working fluids. The freezing phenomenon in PCHE was observed experimentally, and an analytical investigation was carried out to determine how to avoid the freezing phenomenon in PCHE. Finally, a new dimensionless parameter was proposed to analyze the freezing phenomenon. For each PCHE under various operating conditions, the corresponding values were calculated, and the ability to avoid freezing phenomenon was compared quantita-

tively using these values.

2. Experiment apparatus

2.1. Design of PCHEs

Fig. 1 shows the overall and cross-sectional view of PCHE used in this study. Fig. 1(a) shows the fabricated PCHE with the flow direction of the cold channel (blue arrow) and the hot channel (red arrow) indicated. PCHE consists of a total of six plates (two end plates (top and bottom), two cold plates and two hot plates) (Fig. 1(b)). The configuration of the cold channel is designed as a I-type and that of the hot channel plate is set as a N-type. Each plate was assembled using the diffusion bonding method [27] and the material of the plate is STS316L. The 3/8" pipes were welded to PCHE for connection to the test facility. Fig. 1(c) shows the cross-sectional view of PCHE, and it can be seen that the plates are well bonded with no cracks between them. Two tests were conducted under an internal pressure of 40 bar to check for leakage: 1) inspecting the sidewall for bubbles using liquid leak detectors and 2) verifying whether the pressure remained constant after 24 h at 40 bar. Only PCHEs that passed the leak test were used in the experiments.

Fig. 2 shows the geometry of the cross-sectional view and PCHE. A total of four PCHEs with semi-circular channels were made, each with a different flow path diameter (D), pitch (σ), thickness (δ) and number of channels (N). Table 1 lists the detailed specifications of PCHEs. The total heat transfer area was calculated using Eq. (1). Each of four PCHEs was designed to have the same total heat transfer area, although the flow path diameters on the cold and hot channels differ.

$$A_{HT} = \left(\frac{\pi}{2} + 1\right) D_{\text{hot \& cold}} \times L \times N_{\text{channels}} \times N_{\text{plates}} \quad (1)$$

2.2. Experimental setup

For the fluids, N_2 was used as the cold fluid and EGW with a water concentration of 50 wt% was used as the hot fluid. Table 2 presents the physical properties of these fluids. The properties were calculated using the REFPROP (version 10.0) and CoolProp (version 6.4.3) programs [28,29].

Fig. 3(a) shows the schematic of the experimental setup. To reduce heat loss due to convection, PCHE was installed within a vacuum chamber and a vacuum pump was used to maintain a pressure below 50 mTorr. Fig. 3(b) shows the PCHE installation inside the vacuum chamber. To minimize heat loss due to conduction between PCHE and the installation jig, a polyetheretherketone (thermal conductivity: 0.25 W/m·K) block was made to support PCHE exclusively at the vertices of the block.

In the experimental setup, there were three flows: 1) cold flow (blue), 2) hot flow (red), and 3) pre-cooler flow (green). The cold flow (blue) was configured as an open loop. The nitrogen was supplied from the cryogenic liquid nitrogen tank, passing through a heater and cryogenic regulator (B-type; Cash Valve, McKinney, TX, USA) to control the inlet temperature and pressure. A heat insulator, vacuum double piping, and cryogenic valve (C2041-M21; Cryogenic valve, CRYOCOMP, NJ, USA) were used to minimize the heat loss during this process. After passing through PCHE, the cold fluid passed through a mass flow meter (CMF010 Micromotion; Emerson Electric, St Louis, MO, USA) and a mass flow controller (SLA5853; BROOKS INSTRUMENT, PA, USA), respectively, and then discharged to the atmosphere. An additional heater was installed to maintain the working temperature range of the mass flow meter and mass flow controller. The hot flow (red) was configured as a closed loop. A heating circulator (Dyneo DD-600F; Julabo, Seelbach, GER) was used to maintain a constant temperature, and the flow rate was regulated by a magnetic gear pump (DGS2.3; ThrillCrop, Orlando, FL, USA). The flow rate was measured using a mass flow meter (CA003; Oval, Tokyo, JP). The pre-cooler flow (green) was utilized to maintain the cryogenic temperature of the cold fluid. It used nitrogen as the working fluid. The double pipe heat exchanger was

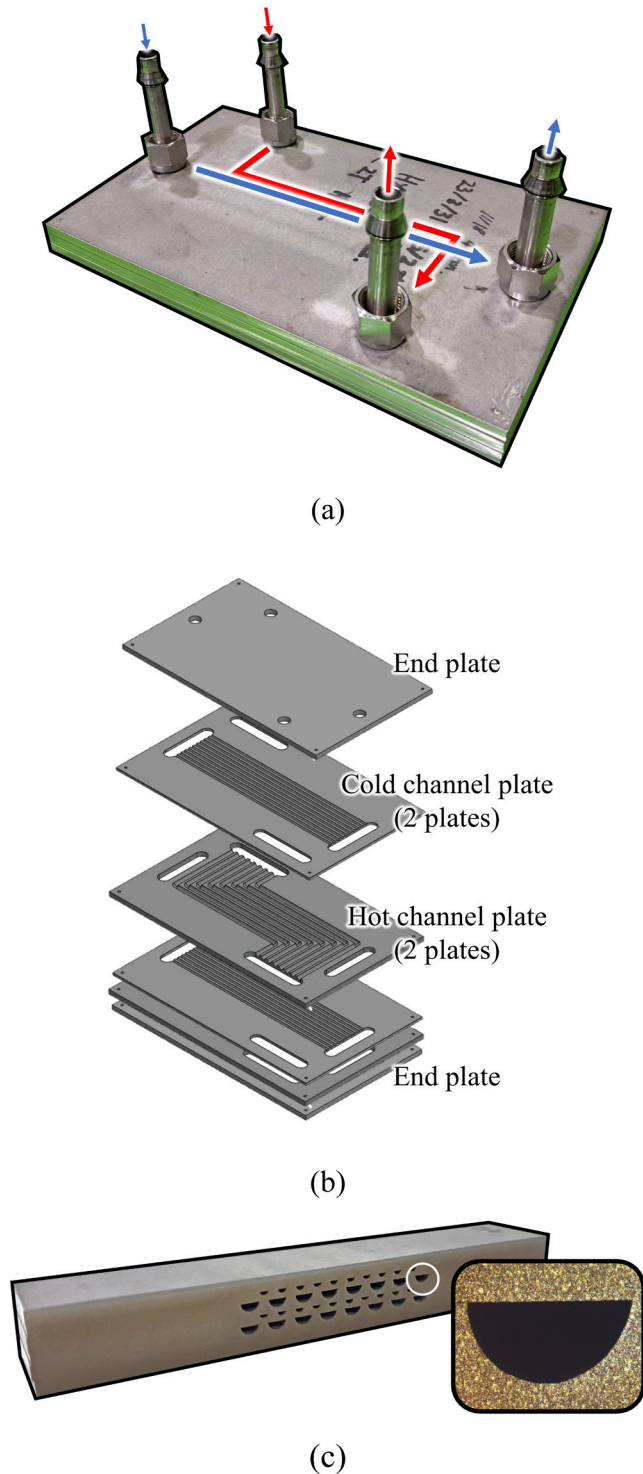


Fig. 1. Overview of PCHE: (a) Fabricated PCHE (red arrow: hot channel, blue arrow: cold channel), (b) Schematic diagram of plates inside PCHE and (c) Cross-sectional view and a microscopic view of the semi-circular channel inside PCHE. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

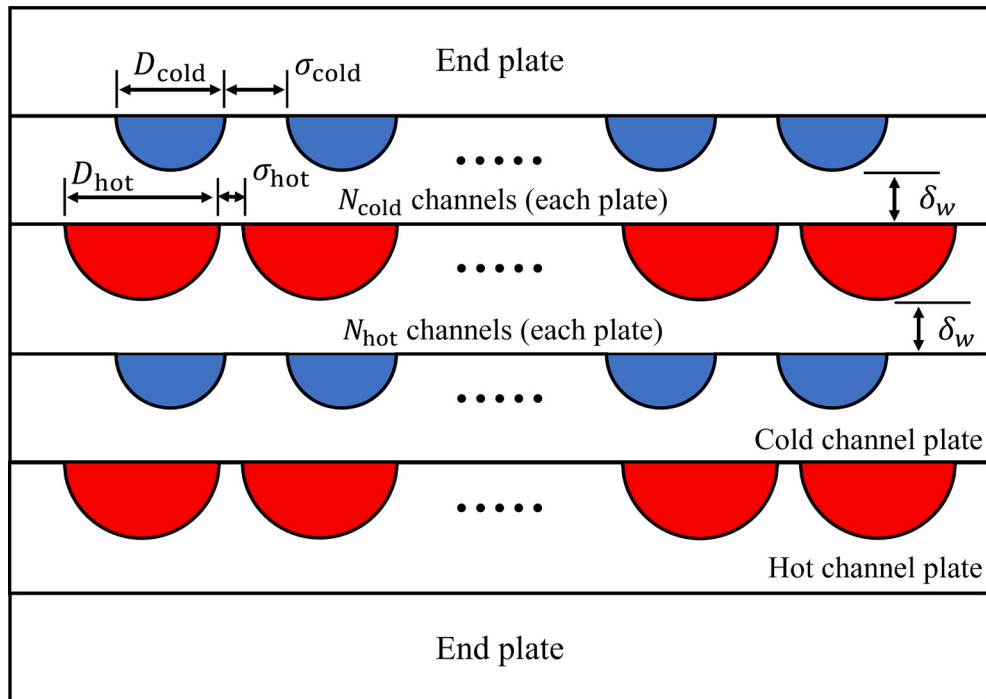


Fig. 2. Configuration of PCHE.

Table 1
Geometries of PCHEs.

	Cold					
	D_{cold} [mm]	N_{channels} [-]	σ_{cold} [mm]	δ_w [mm]	L , channel length [mm]	A_{HT} [mm ²]
HX-01	1	20	0.95	1	160	16,453
HX-02	1	20	0.95	1	160	16,453
HX-03	2	10	2	1	160	16,453
HX-04	2	10	2	1	160	16,453
	Hot					
	D_{hot} [mm]	N_{channels} [-]	σ_{hot} [mm]	δ_w [mm]	L , channel length [mm]	A_{HT} [mm ²]
HX-01	2	16	0.93	1	184	30,274
HX-02	3	11	1.3	1	184	31,220
HX-03	2	16	0.93	1	184	30,274
HX-04	3	11	1.3	1	184	31,220

Table 2
Thermodynamic properties of liquid nitrogen (LN_2) and EGW.

Working fluid	Density [kg/m ³]	Dynamic viscosity [μPa • s]	Thermal conductivity [W/m • K]	Heat capacity [kJ/ kg • K]	Heat of vaporization [kJ/ kg]	Heat of liquefaction [kJ/ kg]
LN_2^*	700.5	80.1	0.103	2.273	165	—
EGW ^{**}	1053.4	2131.4	0.402	3.413	—	226.355 [30]

* 7 bar, −175 °C.

** 1 bar, 40 °C.

installed before the cold fluid entered PCHE. A mass flow controller regulated the flow rate of the pre-cooler line, and an additional PHE and heating circulator were installed before entering the controller to ensure that the mass flow controller operated within the range of the working temperature.

Tables 3–6 present the specifications of the equipment and sensors. The RTD was installed 30 cm away from the inlet and outlet of PCHE and a cube-shaped RTD block was constructed (Fig. 3(b)) to improve the measurement accuracy.

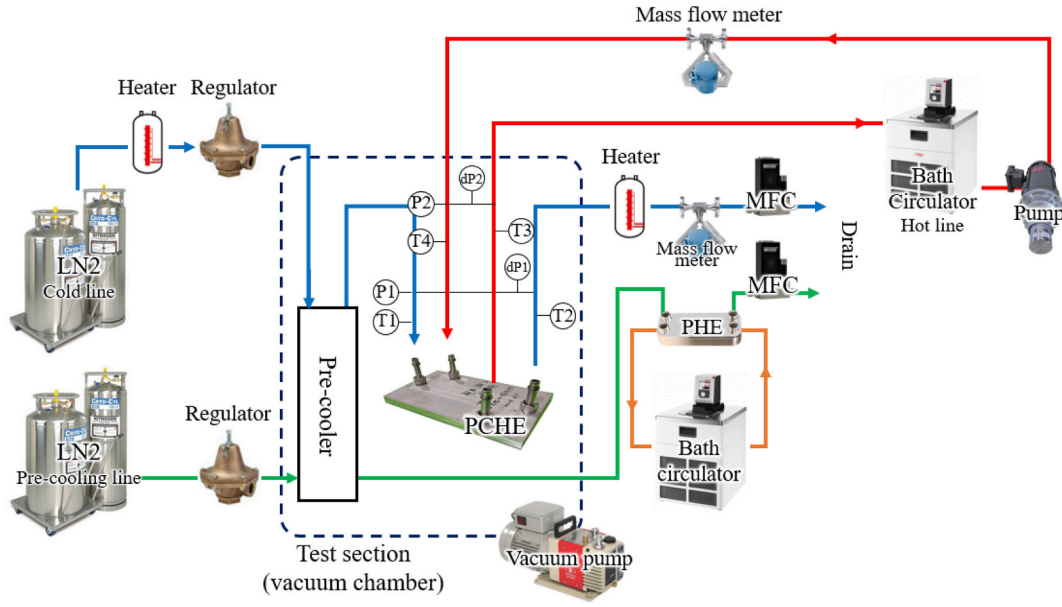
2.3. Experimental conditions

Table 7 lists the experimental conditions. The test was conducted with a co-flow rather than a counter-flow, commonly used to avoid the freezing phenomenon [31]. The flow rate of the cold channel was maintained at 0.2 or 0.1 kg/min, respectively. The inlet temperature of the hot fluid was maintained at 40 °C, and the hot flow rate was decreased from 2.3 kg/min until the differential pressure of the hot channel increased by 20 kPa due to the formation of the solid layer. After the freezing experiment finished, the vacuum chamber was opened to enable PCHE to thaw before proceeding to the next test.

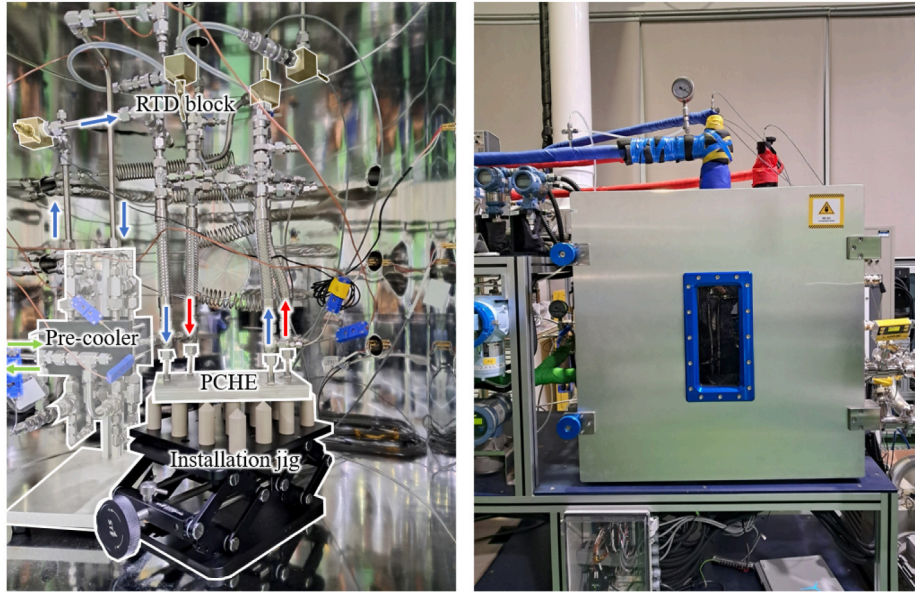
3. Data reduction

3.1. Freezing phenomenon formulation and analysis

Theoretical models describing freezing phenomena and solidifica-



(a)



(b)

Fig. 3. Experimental setup: (a) Schematic diagram of the experimental setup and (b) test section inside the vacuum chamber (left), the outside view of the vacuum chamber (right).

Table 3
PCHE heat transfer test facility component specification.

	Bath circulator	EGW Pump	Regulator	Vacuum pump
Type	—	—	—	Rotary Vane
Manufacturer	Julabo	ThrillCorp	Cash Valve	Woosung Vacuum CO.
Model	DYNEO-DD-600F	—	B-type	MVP60
Note	—	Inverter driven	—	—

tion on the inner wall of a pipe have been widely studied [32–34]. Fig. 4 shows a schematic diagram of the freezing model. The hot fluid enters the pipe of radius r_i with a constant average flow velocity $u_{hot,in}$ and uniform temperature $T_{hot,in}$. The wall temperature T_w is maintained below the freezing point T_f of the hot fluid due to the heat transfer between the cold fluid, which flows parallel to the outer surface of the pipe. $u_{cold,in}$ and h_{cold} represent the inlet velocity of the cold fluid and the heat transfer coefficient at that location, respectively. Assuming that heat transfer begins at $z = 0$, the thickness of the solid layer increases along the z -direction while the radius of the liquid layer r_f decreases. u_{hot} is the mean velocity of the hot fluid at that point, and u_{hot} increases as r_f decreases. T_{cold} and T_{hot} are the mean temperature of the channel at z to $z + dz$, which varies along the z -axis as heat exchanger occurs. T_s is the temperature inside the solid layer and varies along the r direction.

Table 4
PCHE heat transfer test facility component specification for cold channels.

	Temperature		Pressure (Abs)	Pressure (differential)	Flow rate
	Inlet	Outlet			
Type	RTD	RTD	Pressure transmitter	Pressure transmitter	Coriolis
Manufacturer/Model	AccuMac/AM1612	sentech eng/se2304	Rosemount/3051TG	Rosemount/2051CD2	Micromotion/CMF010
Spec	−196 °C–160 °C	—	200 kPa	60 kPa	0.5 kg/min
Accuracy	0.02 °C (at −200 °C)	0.03 °C (at 0 °C)	0.065	0.065	0.1

Table 5
PCHE heat transfer test facility component specification for hot channels.

	Temperature	Pressure (G)	Pressure (differential)	Flow rate
Type	RTD	Pressure transmitter	Pressure transmitter	Coriolis
Manufacturer/Model	sentech eng/se2304	Rosemount/2051CA	Rosemount/2051CD2	Oval/CA003
Spec	—	1000 kPa	60 kPa	2.4 kg/min
Accuracy	0.03 °C (at 0 °C)	0.065	0.065	0.1

Table 6
PCHE heat transfer test facility component specification for the pre-cooler line.

	Bath circulator	PHE
Type	—	—
Manufacturer	DAIHAN Scientific Co	dongil brazing co
Model	SCL-12	DIC 285-19F
Note	—	—

Table 7
Experimental conditions.

	Cold fluid (GN ₂)	Hot fluid (EGW)
Flow configuration	Co-flow	
Inlet temperature [°C]	−166 to −140	40
Inlet mass flow rate [kg/min]	0.2, 0.1	2.3 to 0
Inlet Re	11,794 to 4082	441 to 0

For this study, the following assumptions were made in our model.

- 1) The fluid is incompressible and Newtonian.
- 2) The model is axisymmetric and unsteady state.
- 3) The heat transfer along the z direction is not considered.
- 4) The mass flow rate of the hot channel is constant [35].
- 5) The temperature at the surface of the solid layer, i.e., the interface between the liquid and solid phases, is kept constant at T_f which is the freezing point of the hot fluid [35].
- 6) Peclet number is very large [31–33,35]. Peclet number, which represents the ratio of energy transport by convection to that by conduction, supports this assumption, as convection dominates over conduction in internal forced flows [32].

As the freezing phenomenon occurs, the thickness of the solid layer can be described in terms of both time t and axial position parameter z . The system of governing equations that follow, are stated in non-dimensional form by introducing the following variables and parameters:

$$\begin{aligned}
 R &= r/r_i, \quad Z = z/r_i, \quad \tau = \alpha_s t/r_i^2, \quad \text{Re} = 2ur_i/\nu, \quad \text{Pr} = \nu/\alpha, \quad \alpha = k/\rho C_p \\
 \theta_s &= (T_f - T_s)/(T_f - T_{\text{cold}}), \quad \theta = (T - T_f)/(T_{\text{hot,in}} - T_f) \\
 \text{Ste}_s &= (T_f - T_{\text{cold}})C_{p,s}/h_{fs}, \quad \text{Ste}_l = (T_{\text{hot,in}} - T_f)C_{p,l}/h_{ls} \\
 \text{Bi} &= \frac{hr_i}{k_s}, \quad \text{Pe} = \frac{2r_i u}{\alpha} = \text{Re} \cdot \text{Pr}
 \end{aligned} \quad (2)$$

R and Z denote the distance dimensionless numbers in the radial and axial directions, respectively. r_i is the initial radius of the pipe. τ is the non-dimensional time and it can be obtained using the thermal diffusivity of the solid layer (α_s), time (t) and the inner radius (r_i). Reynolds number (Re) is the product of the flow velocity (u) and the inner

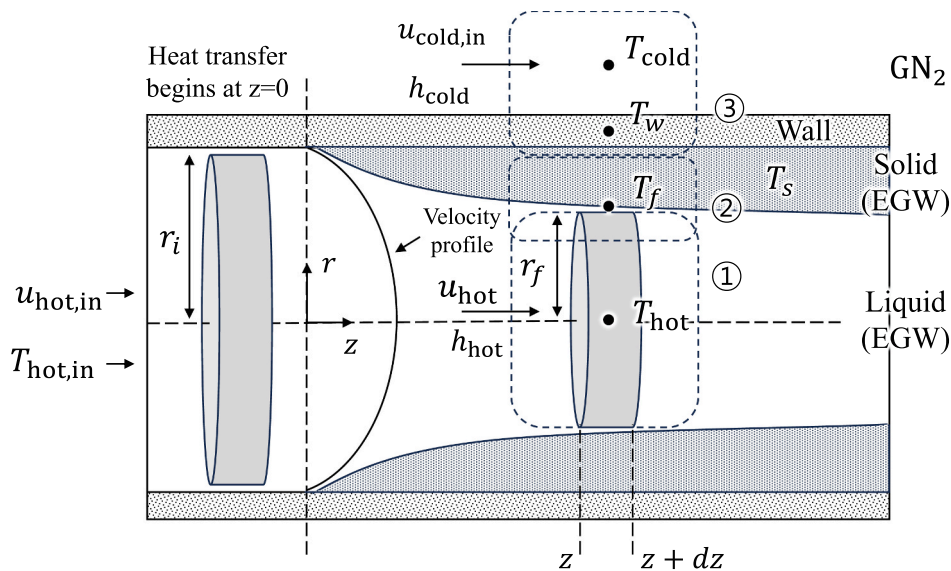


Fig. 4. Schematic diagram for freezing phenomenon inside a pipe with a fluid flow (① Liquid section ② Liquid-solid interface ③ solid-wall section).

diameter ($2r_i$) divided by the kinematic viscosity coefficient (ν). Prandtl number (Pr) can be obtained using the kinematic viscosity (ν) divided by the thermal diffusivity (α), where α is a function of thermal conductivity (k), density (ρ) and specific heat (C_p). θ_s and θ are the non-dimensional temperature of the solid layer and the temperature of the hot liquid. Each dimensionless values can be derived from the freezing point (T_f), the temperature of the cold channel (T_{cold}), and the inlet temperature of the hot channel ($T_{\text{hot, in}}$). Ste is Stefan number defined as the ratio of the sensible heat to the heat of liquefaction. It can be obtained using the specific heat (C_p) and the heat of liquefaction of the fluid (h_{ls}) respectively. Biot number (Bi) is a ratio of the thermal resistance for the conduction of a solid to the resistance for the convection at the interface. Finally, Peclet number (Pe) can be expressed as a function of the radius (r_i), flow velocity (u) and thermal diffusivity (α) or simply as Reynolds number (Re) multiplied by Prandtl number (Pr). Nusselt number (Nu) correlation for both the hot and cold fluids is given as follows:

$$\text{Nu}_{\text{hot}} = 4.089, \text{Nu}_{\text{cold}} = 0.023\text{Re}_{\text{cold}}^{0.8}\text{Pr}_{\text{cold}}^{0.4} \quad (3)$$

Nu_{hot} , a constant value of 4.089, was used because the hot fluid maintains the laminar flow under the experimental conditions [37]. Nu_{cold} was calculated using the Dittus Boelter correlation because the cold fluid maintains the turbulent flow.

The governing equations for the freezing phenomenon inside the channel can be derived from the conservation laws of mass, momentum, and energy. These governing equations are shown in Fig. 4 for each of the three regions.

- 1) liquid region ($0 < r < r_f, 0 < R < R_f$)
- 2) Phase interface between the liquid and the solid layer ($r = r_f, R = R_f$)
- 3) Inner section of the solid layer with the boundary condition of the heat flux from the cold fluid ($r_f \leq r < 1, R_f \leq R < 1$),

First, in the liquid region ($0 < r < r_f, 0 < R < R_f$), assuming that the loss of energy in the unit volume and the heat transfer due to convection are equal, the energy can be expressed as follows:

$$C_{p,l}(\rho_{\text{hot}}u_{\text{hot}}\pi r_f^2 dz) \frac{dT_{\text{hot}}}{dz} = h_{\text{hot}} \cdot 2\pi r_f dz \cdot (T_{\text{hot}} - T_f) \quad (4)$$

$$\text{Pe} \frac{d\theta_{\text{hot}}}{dz} + 2\text{Nu}_{\text{hot}}\theta_{\text{hot}} = 0 \quad (4a)$$

The right-hand side of Eq. (4) expresses the energy loss and is a product of the specific heat ($C_{p,l}$), the unit volume and the temperature gradient (dT_{hot}/dz). The mean velocity u_{hot} can be replaced by $(r_i/r_f)^2 u_{\text{hot, in}}$ and finally, Eq. (4) can be re-written in non-dimensional form as Eq. (4a).

Second, for phase interface between the liquid and solid layer ($r = r_f, R = R_f$), Eqs. (5), (5a) and (5b) can be expressed as follows:

$$h_{\text{ls}}\rho_{\text{hot}} \frac{\partial r_f}{\partial t} = h_{\text{hot}}(T_{\text{hot}} - T_f) + k_s \left(\frac{\partial T_s}{\partial r} \right)_{r=r_f} \quad (5)$$

$$\frac{\partial R_f}{\partial \tau} = \frac{(T_f - T_{\text{cold}})C_{p,s}}{h_{\text{ls}}} \left\{ -\frac{\partial \theta_s}{\partial R} + \frac{k_l}{k_s} \frac{T_{\text{hot, in}} - T_f}{T_f - T_{\text{cold}}} \frac{1}{R_f} \text{Nu}_{\text{hot}} \cdot \theta_{\text{hot}} \right\} \quad (5a)$$

$$\left(\frac{\partial R_f}{\partial \tau} \right) = -\text{Ste}_s \left(\frac{\partial \theta_s}{\partial R} \right)_{R=R_f} + \left(\frac{\phi \theta_{\text{hot}}}{R_f} \right) \quad (5b)$$

$$\phi(R_f) = \left(\frac{C_p}{2k} \right)_{\text{sl}} \text{Ste}_l, \text{Nu}_{\text{hot}}(R_f)$$

The right-hand side of Eq. (5) represents the heat flux due to the convection and conduction at the interface. The heat flux difference between the two terms contributes to the growth rate of the solid layer. It can be expressed in terms of the heat of solidification (h_{ls}), the density

(ρ_{hot}) and the growth rate of the solid layer ($\partial r_f/\partial t$) as shown on the left-hand side of Eq. (5). Similarly, Eq. (5) can be expressed in non-dimensional form as Eq. (5b) using R_f , θ_s , and θ_b . Eq. (5a) is obtained by converting Eq. (5) to Eq. (5b). On the right-hand side of Eq. (5a), the term $k_l(T_{\text{hot, in}} - T_f)/k_s(T_f - T_{\text{cold}})$ is observed, which is referred to as the superheat number (Su) and represents the ratio between the sensible heat in the liquid and solid. The Su number is the main variable that determines the shape of the solid layer [32,33,36].

Finally, the inner section of the solid layer ($r_f \leq r < 1, R_f \leq R < 1$) can be represented by a heat conduction equation with a cylindrical coordinate system. Both the heat conduction equation and boundary conditions are given below:

$$\left(\frac{\partial^2 \theta_s}{\partial R^2} \right) + \frac{1}{R} \left(\frac{\partial \theta_s}{\partial R} \right) = 0, (R_f \leq R < 1) \quad (6)$$

$$\theta_s = 0, (\text{b.c} : R = R_f) \quad (6a)$$

$$\left(\frac{dT_s}{dr} \right)_{r=r_i} = \frac{h_{\text{tot, cold}}}{k_s} (-T_{\text{cold}} + T_s) \quad (6b)$$

$$\left(\frac{\partial \theta_s}{\partial R} \right)_{R=1} = \text{Bi}[1 - \theta_s], (\text{b.c} : R = 1)$$

$$\frac{1}{h_{\text{tot, cold}}} = \left[\frac{\delta_w A_i}{k_w A_{HT}} + \frac{A_i}{h_{\text{cold}} A_{HT}} \right] \quad (7)$$

$h_{\text{tot, cold}}$ in Eq. (6b) can be expressed as a term for the thermal resistance at the tube wall and the convective heat transfer coefficient of the cryogenic fluid, as shown on the right-hand side of Eq. (7). Assuming that the thickness of the tube wall (δ_w) is small enough (1 mm) and the thermal conductivity (k_w) of STS316L, which forms PCHE, is large enough ($16.3 \text{ W/m} \cdot \text{K}$), the thermal resistance of the tube wall (10^{-4}) is negligible compared to the convective heat transfer coefficient term (10^{-2}) and $h_{\text{tot, cold}} = h_{\text{cold}}$.

Solving the differential equation Eq. (6) with the given boundary conditions gives a solution as Eq. (8) that represents the temperature gradient (θ_s) inside the solid layer.

$$\theta_s = \left(\frac{\text{Bi}}{1 - \text{Bi} \cdot \ln R_f} \right) \ln \left(\frac{R}{R_f} \right) \quad (8)$$

In a previous study [34], the decreasing rate of R_f from the initial time ($t \rightarrow 0$) was used to determine whether a freeze had occurred. Substituting Eq. (8) into Eq. (5b) gives Eq. (9) as follows:

$$R_f \left(\frac{\partial R_f}{\partial \tau} \right) = -\text{Ste}_s \left(\frac{\text{Bi}}{1 - \text{Bi} \cdot \ln R_f} \right) + \phi \theta \quad (9)$$

when $\tau \rightarrow 0$, θ and R_f are 1 by the definition of the non-dimensional terms. Thus, the initial growth rate of R_f can be expressed as in Eq. (10) and Eq. (10a).

$$\left(\frac{\partial R_f}{\partial \tau} \right)_{\tau \rightarrow 0} = -\text{Ste}_s \text{Bi} + \phi \quad (10)$$

$$= -\frac{(T_f - T_{\text{cold}})C_{p,s}}{h_{\text{ls}}} \text{Bi} + \frac{1}{2} \frac{C_{p,s}}{C_{p,l}} \frac{k_l}{k_s} \frac{(T_{\text{hot, in}} - T_f)C_{p,l}}{h_{\text{ls}}} \text{Nu}_{\text{hot}} \quad (10a)$$

Depending on the derivative term ($\partial R_f/\partial \tau|_{\tau \rightarrow 0}$) on the left side of Eq. (10), there are three different situations to consider: 1) If the derivative is less than zero, the solid layer is formed (freezing phenomenon occurs), 2) if it is equal to zero, it indicates a steady state, and 3) if it is greater than zero, the thickness of the solid layer will decreased (melting occurs). Eq. (10) can be rewritten as Eq. (10a) to examine the effects of the properties of the hot and cold fluid on $\partial R_f/\partial \tau|_{\tau \rightarrow 0}$. To make $\partial R_f/\partial \tau|_{\tau \rightarrow 0}$ a negative value (freezing phenomenon occurs) T_{cold} , $T_{\text{hot, in}}$ and Nu_{hot} needs to be decreased, or Bi (as a function of h_{cold}) should be increased.

In this study, Bi is defined as Bi_{freeze} , where the value of $\partial R_f / \partial \tau|_{\tau \rightarrow 0}$ satisfies zero, as shown in Eq. (11), and proposed it as a standard for the occurrence of the freezing phenomenon.

$$Bi_{freeze} = \frac{1}{2} \frac{k_l}{k_s} \frac{(T_{hot,in} - T_f) Nu_{hot}}{(T_f - T_{cold})} \quad (11)$$

$$Bi = \frac{h_{cold} r_i}{k_s} \quad (11a)$$

Any point inside cold channel Bi can be calculated as Eq. (11a) and if this Bi value is greater than Bi_{freeze} , the derivative $\partial R_f / \partial \tau|_{\tau \rightarrow 0}$ will be negative, which means that freezing phenomenon has begun. On the other hand, if the Bi_{freeze} is calculated to be negative, it means that the temperature of the cold fluid (T_{cold}) is higher than the freezing point of the hot fluid (T_f). This means that the freezing phenomenon cannot occur.

3.2. Heat transfer model and thermal model along the channel

Eqs. (12) and (13) give the heat transfer rate of the cold channel and the hot channel.

$$Q_{cold} = \dot{m}_{cold} (i_{cold,in} - i_{cold,out}) \quad (12)$$

$$Q_{hot} = \dot{m}_{hot} (i_{hot,out} - i_{hot,in}) \quad (13)$$

The enthalpy was calculated using the measured temperatures of the RTDs installed at the inlet and outlet of the flow passages, and the values of the differential pressure. The values of each data were used when the steady state was maintained for more than 20 min.

The overall heat transfer coefficient U can be obtained by using Eqs. (14) and (14a).

$$U = \frac{Q_{cold}}{A_{HT} \cdot \Delta T_{LMTD}} \quad (14)$$

$$\Delta T_{LMTD} = \frac{(T_{hot,in} - T_{cold,out}) - (T_{hot,out} - T_{cold,in})}{\ln \left(\frac{T_{hot,in} - T_{cold,out}}{T_{hot,out} - T_{cold,in}} \right)} \quad (14a)$$

Eq. (14) uses the value of Q_{cold} because there is no phase change in the cold channel. Log mean temperature degree ΔT_{LMTD} for the parallel flow was calculated as in Eq. (14a).

To calculate the Bi/Bi_{freeze} value, the information of T_{cold} and T_{hot}

along the z direction inside the channel is needed. The temperature at each location was calculated using the ϵ -NTU method. The ϵ -NTU method is useful when the inlet and outlet temperatures of the heat exchanger are known. Previous studies [23,38] have also used the ϵ -NTU method to obtain the internal temperature distribution of PCHEs.

The PCHE internal flow path is simplified as a 1D parallel flow heat exchanger to apply the ϵ -NTU method. Fig. 5 shows the schematic diagram divided into N sections.

The following procedure is used to find the outlet temperature $T_{cold,n+1}$, $T_{hot,n+1}$ in the n th interval. In the n th interval, the NTU can be obtained as Eq. (15).

$$NTU_n = \frac{UA_{HT,n}}{\min(C_{cold,n}, C_{hot,n})} \quad (15)$$

$$C_{cold,n} = \dot{m}_{cold,in} \cdot C_{p,cold}, \quad C_{hot,n} = \dot{m}_{hot,in} \cdot C_{p,hot} \quad (15a)$$

To calculate $UA_{HT,n}$, UA_{HT} was obtained from Eq. (14) and divided by N , assuming that U is uniform for all locations. $C_{cold,n}$ and $C_{hot,n}$ are the heat capacity calculated from Eq. (15a) as a function of the mass flow rate and specific heat.

Heat transfer effectiveness (ϵ_n) for the parallel flow can be derived from Eq. (16)[39].

$$\epsilon_n = \frac{1 - \exp[-NTU_n(1 + c_n)]}{1 + c_n} \quad (16)$$

$$c_n = \frac{\min(C_{cold,n}, C_{hot,n})}{\max(C_{cold,n}, C_{hot,n})} \quad (16a)$$

$$Q_n = \epsilon_n Q_{max,n} \quad (17)$$

$$Q_{max,n} = \min(C_{cold,n}, C_{hot,n}) \cdot (T_{hot,n} - T_{cold,n}) \quad (17a)$$

$$T_{cold,n+1} = T_{cold,n} + Q_n / C_{cold,n}$$

$$T_{hot,n+1} = T_{hot,n} - Q_n / C_{hot,n} \quad (18)$$

In the right-hand side of Eq. (16), the heat capacity ratio (c_n) is the ratio of the minimum and maximum values between the heat capacity rates of the cold and hot fluid and can be expressed as in Eq. (16a). This allows the heat transfer rate at n th interval (Q_n) to be given by Eq. (17) as a product of the effectiveness (ϵ_n) and $Q_{max,n}$ which calculated from Eq. (17a). Finally, the exit temperatures $T_{cold,n+1}$, $T_{hot,n+1}$ are obtained from Eq. (18).

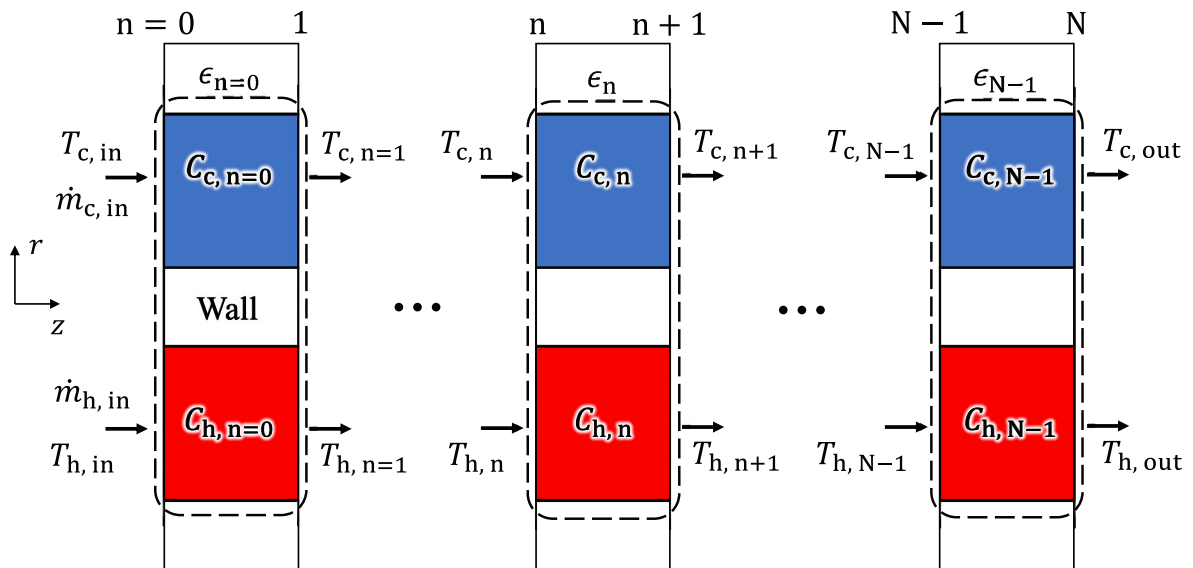


Fig. 5. Schematic diagram of discretization along the flow channels.

4. Results and discussion

4.1. Thermal and hydraulic performance of PCHEs

To analyze the thermal performance of the heat exchanger, heat transfer experiments were conducted within a range of flow rate conditions where the freezing phenomenon does not occur. Fig. 6 shows the heat transfer rate of the cold and hot channels (Q_{cold} , Q_{hot}), respectively, which were derived from the Eqs. (12) and (13). Reynolds numbers obtained from the properties calculated from the mean values of the inlet and outlet temperatures ranged from 4000 to 10500 on the cold fluid and 430–447 on the hot fluid. PCHE could exchange about 167–575 W of heat. The error of the heat exchange amount between the hot and cold fluids was within $\pm 10\%$.

As described in Section 2.3, the experiments were performed by lowering the flow rate of the hot channel while keeping the flow rate of the cold fluid constant to observe the freezing phenomenon. One method of observing the freezing phenomenon is to examine the pressure of the hot channel [40]. When the freezing phenomenon begins, the cross-sectional area of the hot channel gradually decreases, and the differential pressure increases as the flow velocity increases. This behavior was observed for all four PCHEs tested in the experiment. Fig. 7 shows the relationship between the mass flow rate and the differential pressure of the hot channel for HX-04. The inlet and outlet temperatures of the cold and hot channel are listed in Table 8.

In Fig. 7, when $\dot{m}_{\text{hot}}$ decreased from 2 to 0.25 kg/min, differential pressure (ΔP_{hot}) also decreased. This trend changes when $\dot{m}_{\text{hot}}$ is in the range of 0.25 to 0.15 kg/min as ΔP_{hot} begins to increase. The reason for this is that a larger flow rate on the cold fluid increases the convective heat transfer coefficient, which results in a higher heat flux and consequently, a higher probability of the freezing phenomenon.

The evolution of the differential pressure and flow rate over time at points A and B as shown in Fig. 7, are plotted in Fig. 8. In Fig. 8(a), it can be seen that the flow and differential pressure remain constant because no freezing phenomenon occurs. On the other hand, in Fig. 8(b), the differential pressure increases, and the flow rate decreases as the freezing phenomenon occurs, until the channel is completely blocked.

In our previous work [26], freezing experiments were conducted and a contour map of the differential pressure of the hot channel was proposed with respect to the mass flow rates of the cold and hot channels. This contour map was used to show the operating conditions where the possibility of the freezing phenomenon is high. Although, this method of observing differential pressure has limitations in that it does not reflect the physical properties of the fluid, inlet conditions, geometry of the channel, etc. Therefore, in order to generalize the freezing phenomenon in PCHE, dimensionless number that represents the freezing phenomenon is required.

In this study, $\text{Bi}/\text{Bi}_{\text{freeze}}$ is proposed as a dimensionless number that

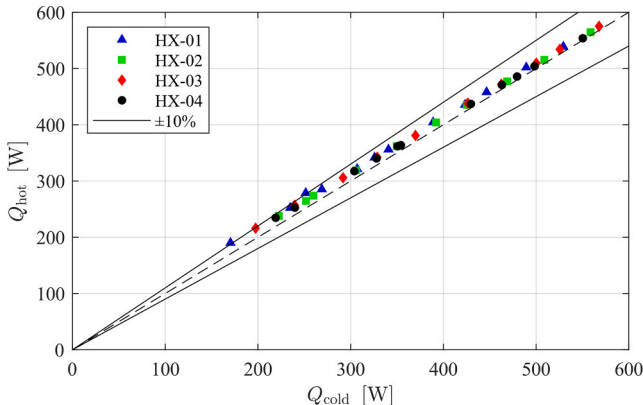


Fig. 6. Heat transfer rate of PCHEs between the hot fluid and cold channel.

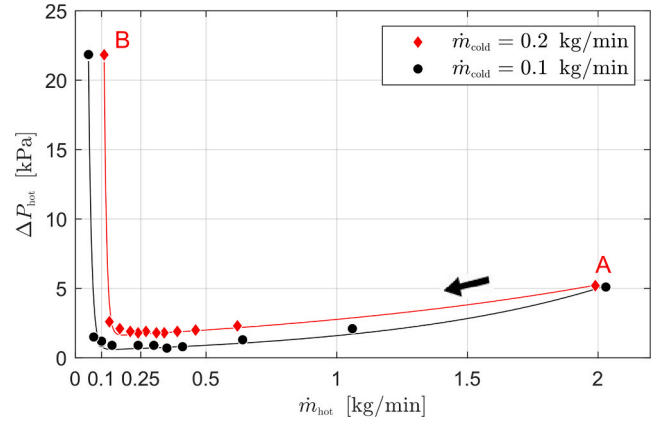


Fig. 7. Relationship between the ΔP_{hot} and the $\dot{m}_{\text{hot}}$ (HX-04).

Table 8

Inlet and outlet temperature of both cold and hot channel (HX-04).

	$\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$	$\dot{m}_{\text{cold}} = 0.1 \text{ kg/min}$
$T_{\text{cold,in}} [^{\circ}\text{C}]$	−153 to −143	−157 to −149
$T_{\text{cold,out}} [^{\circ}\text{C}]$	−57 to 8	−56 to 18
$T_{\text{hot,in}} [^{\circ}\text{C}]$	40	40
$T_{\text{hot,out}} [^{\circ}\text{C}]$	2 to 35	−14 to 37

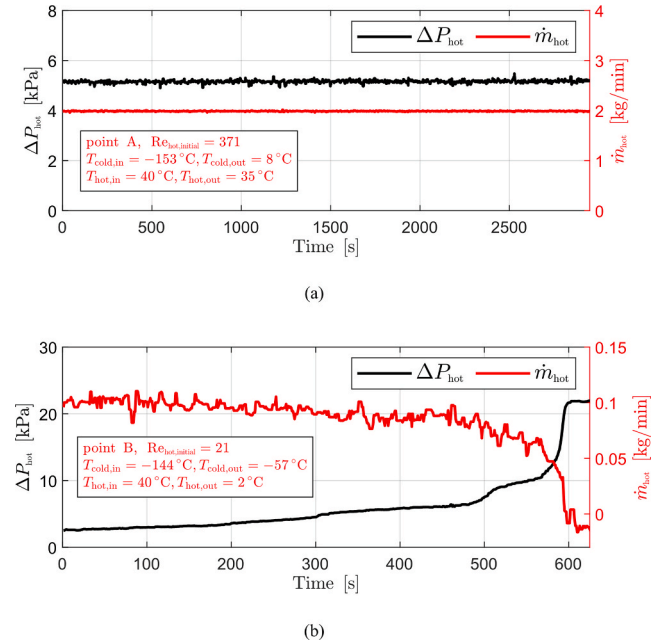


Fig. 8. Temporal data on the differential pressure and mass flow rates of the hot channel (at $\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$). (a) Normal condition without the freezing phenomenon at point A and (b) freezing condition at point B (HX-04).

can be used to determine the occurrence of the freezing phenomenon. $\text{Bi}/\text{Bi}_{\text{freeze}}$ was calculated at each position within the hot channel. If the obtained value is negative, it indicates that the denominator term $\text{Bi}_{\text{freeze}}$ is negative, meaning that there is no possibility of the freezing phenomenon at that point. Based on the experimental data, contour maps for $\text{Bi}/\text{Bi}_{\text{freeze}}$ were plotted, and the freezing phenomenon was investigated by monitoring the regions with negative $\text{Bi}/\text{Bi}_{\text{freeze}}$ values.

4.2. Comparison of the freezing phenomenon with different cold flow rates

Fig. 9 shows the contour map of Bi/Bi_{freeze} for different cold flow rates with a constant channel diameter for the hot channel. On the contour map, arrows indicate the direction of time and position. The position is represented as dimensionless number $l = L/L_{channel}$. For time, it flows from top to bottom, and for position, the left side of the contour represents the entrance, and the right side represents the exit. In each case, with the $\dot{m}_{cold}$ held constant, the experiment was conducted by decreasing the $\dot{m}_{hot}$. After reducing the $\dot{m}_{hot}$, the cold and hot outlet temperatures were monitored until they stabilized (fluctuated within ± 0.1 °C for 20 min). Once the temperature stabilized the data were recorded after a period of 10–15 min. Then, $\dot{m}_{hot}$ was reduced again, and the same process was repeated. Inlet and outlet temperatures for the cold and hot channels in each contour map shown in Fig. 9 are listed in Table 9.

The Bi/Bi_{freeze} value decreases as it approaches the exit and then becomes negative. The point where Bi/Bi_{freeze} becomes zero is marked with a white dotted line on the contour map. To the right of the white dotted line is the region where Bi/Bi_{freeze} has a negative value, so the freezing phenomenon cannot occur. In this study, this region is denoted as a Freeze Safety Region (FSR). The size of this region is compared to determine whether the operating conditions are suitable for avoiding the freezing phenomenon. If the Freeze safety region is large, the possibility of the freezing phenomenon in the channel is low. In contrast, the freezing phenomenon will occur if it is small or non-existent. Fig. 9 shows that under the condition in which $\dot{m}_{cold}$ remains constant, the size of the FSR continues to decrease and then disappears as $\dot{m}_{hot}$ decreases. In other word, the smaller $\dot{m}_{hot}$ is unsuitable for avoiding the freezing phenomenon. To analyze the effect of the mass flow rate of the cold channel, similar Re_{hot} cases were compared: as the $\dot{m}_{cold}$ decreased, the FSR increased, which is relatively suitable for avoiding the freezing phenomenon. The same conclusion can be drawn from the freezing analysis inferred from the differential pressure graph in Fig. 7.

The contour maps (Left: $\dot{m}_{cold} = 0.2$ kg/min, $\dot{m}_{hot,initial} = 0.11$ kg/min & Right: $\dot{m}_{cold} = 0.1$ kg/min, $\dot{m}_{hot,initial} = 0.05$ kg/min) at the bottom of Fig. 9 represent the case where the hot fluid channel is completely blocked by a solid layer, and no heat transfer occurs. This is the same situation as in Fig. 8(b). At this point, T_{cold} decreases, causing Bi_{freeze} , calculated using Eq. (11), to decrease as well. As a result,

Table 9

Inlet and outlet temperature of both cold and hot channel in each contour map shown in Fig. 9 (HX-04).

$\dot{m}_{cold} = 0.2$ kg/min				$\dot{m}_{cold} = 0.1$ kg/min			
$\dot{m}_{hot} = 0.39$ kg/min		$\dot{m}_{hot} = 0.41$ kg/min		$\dot{m}_{hot} = 0.24$ kg/min		$\dot{m}_{hot} = 0.24$ kg/min	
$T_{cold,in}$ [°C]	−145	$T_{cold,out}$ [°C]	−27	$T_{cold,in}$ [°C]	−153	$T_{cold,out}$ [°C]	8
$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	22	$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	26
$\dot{m}_{hot} = 0.24$ kg/min				$\dot{m}_{hot} = 0.24$ kg/min			
$T_{cold,in}$ [°C]	−144	$T_{cold,out}$ [°C]	−38	$T_{cold,in}$ [°C]	−152	$T_{cold,out}$ [°C]	−7
$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	14	$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	19
$\dot{m}_{hot,initial} = 0.11$ kg/min				$\dot{m}_{hot,initial} = 0.05$ kg/min			
$T_{cold,in}$ [°C]	−143	$T_{cold,out}$ [°C]	−52	$T_{cold,in}$ [°C]	−149	$T_{cold,out}$ [°C]	−39
$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	2	$T_{hot,in}$ [°C]	40	$T_{hot,out}$ [°C]	−8

Bi/Bi_{freeze} increases throughout the entire channel.

4.3. Comparison of the freezing phenomenon with different sizes of the hot channel

This section examines the influence of the diameter of the hot channel on the freezing phenomenon, using the same cold channel. Fig. 10 shows the contour map for Bi/Bi_{freeze} of HX-03 ($D_{cold} : 2$ mm, $D_{hot} : 2$ mm), and HX-04 ($D_{cold} : 2$ mm, $D_{hot} : 3$ mm) at a mass flow rate of 0.2 kg/min for the cold channel. It can be observed that as the size of hot channel increases, the FSR also increases, indicating a lower probability of the freezing phenomenon. The inlet and outlet temperatures for the cold and hot channels in each contour map (HX-03 and HX-04) shown in Fig. 10 are listed in Table 10.

4.4. Guidelines to avoid freezing phenomenon in PCHEs using FSR

Fig. 11 illustrates FSR and ΔP_{hot} curves of HX-04 ($D_{cold} : 2$ mm, $D_{hot} : 3$ mm) as a function of Re_{hot} . The y-axis of the graph represents the dimensionless number l . The gray shaded area indicates the FSR with l_{FSR} denoting the size of FSR at a specific Re_{hot} . The FSR curves in Fig. 11 gives following information: 1) as the mass flow rate of cold channel

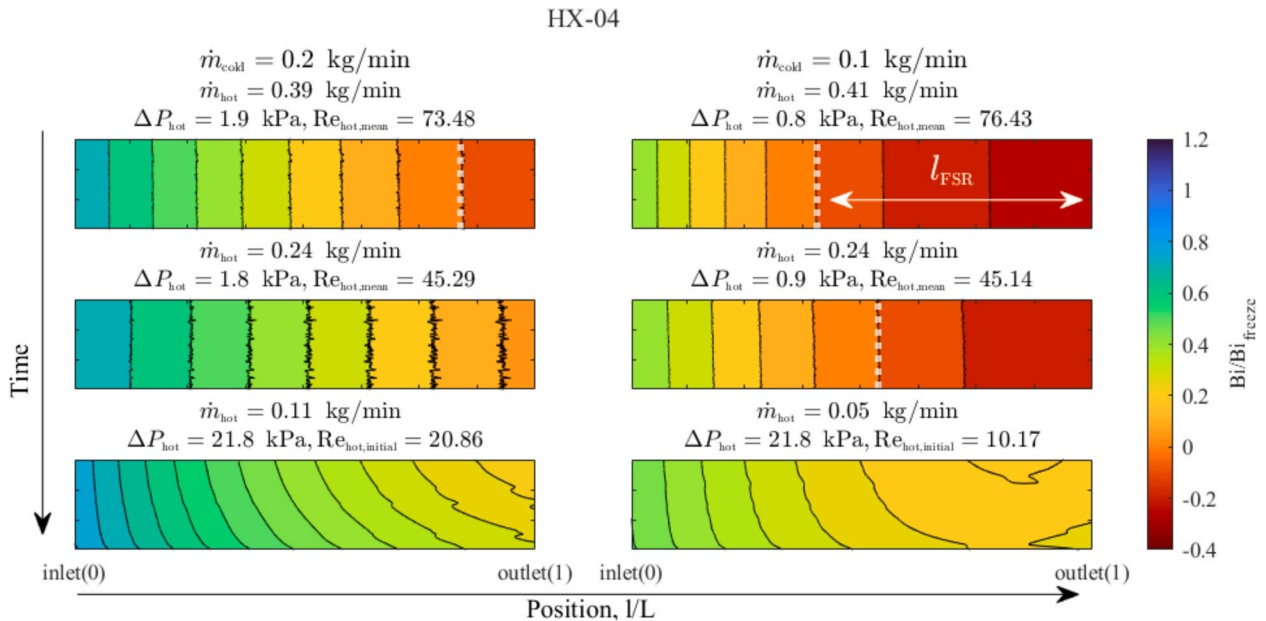


Fig. 9. Bi/Bi_{freeze} contour of HX-04 with different mass flow rates: $\dot{m}_{cold} = 0.2$ kg/min (left) and $\dot{m}_{cold} = 0.1$ kg/min (Right).

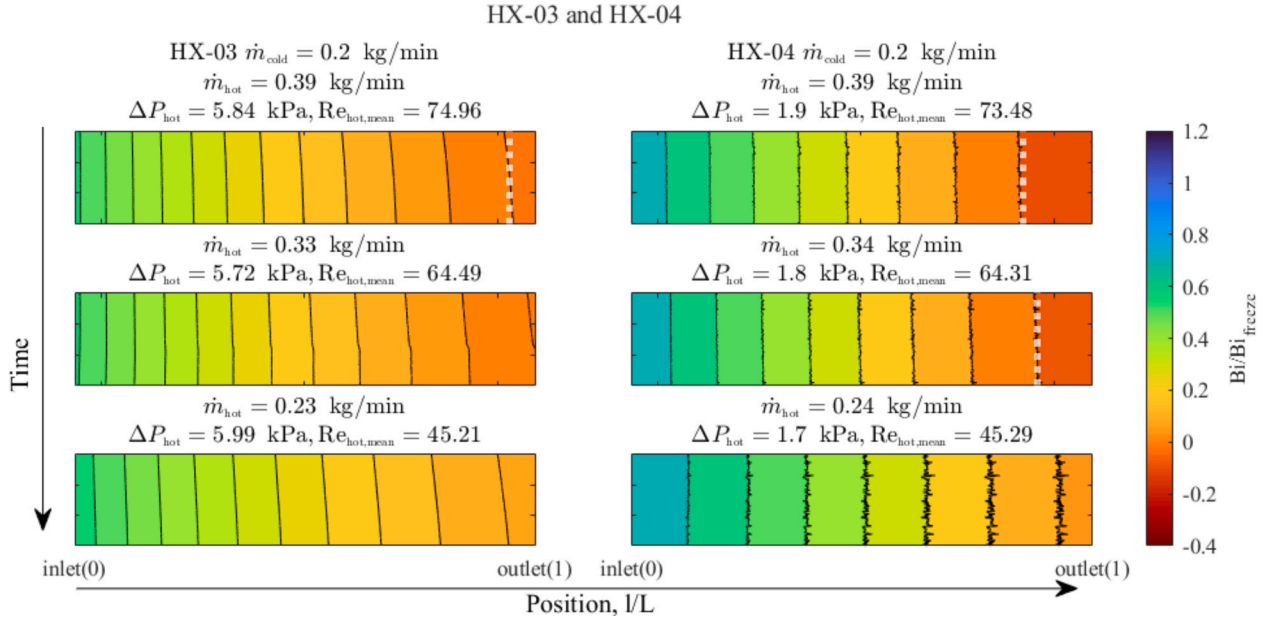


Fig. 10. Bi/Bi_{freeze} contour of HX-03 and HX-04 with different mass flow rates (Left) HX-03 ($D_{\text{cold}} : 2 \text{ mm}$, $D_{\text{hot}} : 2 \text{ mm}$) and (Right) HX-04 ($D_{\text{cold}} : 2 \text{ mm}$, $D_{\text{hot}} : 3 \text{ mm}$).

increases, and 2) as the flow rate on the hot channel decreases (i.e., Re_{hot} decreases), the size of FSR becomes smaller making it vulnerable to freezing phenomenon. This trend is similar with the ΔP_{hot} curves as described in section 4.1.

As the FSR curves gives the value of l_{FSR} , the ability of freezing avoidance for each mass flow rate of cold fluid can be compared quantitatively. For example, when Re_{hot} is 100, the l_{FSR} is 0.28 at $\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$ and 0.59 at $\dot{m}_{\text{cold}} = 0.1 \text{ kg/min}$. The size of FSR increased by 2.1 times as $\dot{m}_{\text{cold}}$ is reduced by half. This indicates that at the operating condition of $\dot{m}_{\text{cold}} = 0.1 \text{ kg/min}$, the ability to avoid freezing phenomenon is 2.1 times greater.

Fig. 12 shows the FSR curve for all PCHEs used in the freezing experiments. All PCHEs exhibit a similar trend. The graph indicates that as Re_{hot} decreases, PCHE with a larger diameter of hot channel has a larger the FSR, making it more effective at preventing the freezing phenomenon. For example, when Re_{hot} is 100, the l_{FSR} of the HX-01 ($D_{\text{cold}} : 1 \text{ mm}$, $D_{\text{hot}} : 2 \text{ mm}$) is 0.70 and 0.81 in case of HX-02 ($D_{\text{cold}} : 1 \text{ mm}$, $D_{\text{hot}} : 3 \text{ mm}$). In contrast, as Re_{hot} increases, the size of the FSR becomes similar regardless of PCHE's hot channel diameter. This result indicates that the diameter of the hot channel has a more critical effect on the freezing phenomenon when $Re_{\text{hot,mean}}$ is small.

Table 10

Inlet and outlet temperature of both cold and hot channel in each contour map shown in Fig. 10 (HX-03 and HX-04).

HX-03, $\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$				HX-04, $\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$			
$\dot{m}_{\text{hot}} = 0.39 \text{ kg/min}$				$\dot{m}_{\text{hot}} = 0.39 \text{ kg/min}$			
$T_{\text{cold,in}}$	−154	$T_{\text{cold,out}}$	−34	$T_{\text{cold,in}}$	−145	$T_{\text{cold,out}}$	−27
[°C]		[°C]		[°C]		[°C]	
$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	21	$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	22
[°C]		[°C]		[°C]		[°C]	
$\dot{m}_{\text{hot}} = 0.33 \text{ kg/min}$				$\dot{m}_{\text{hot}} = 0.34 \text{ kg/min}$			
$T_{\text{cold,in}}$	−153	$T_{\text{cold,out}}$	−38	$T_{\text{cold,in}}$	−145	$T_{\text{cold,out}}$	−29
[°C]		[°C]		[°C]		[°C]	
$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	19	$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	19
[°C]		[°C]		[°C]		[°C]	
$\dot{m}_{\text{hot}} = 0.23 \text{ kg/min}$				$\dot{m}_{\text{hot}} = 0.24 \text{ kg/min}$			
$T_{\text{cold,in}}$	−156	$T_{\text{cold,out}}$	−48	$T_{\text{cold,in}}$	−144	$T_{\text{cold,out}}$	−38
[°C]		[°C]		[°C]		[°C]	
$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	13	$T_{\text{hot,in}}$	40	$T_{\text{hot,out}}$	14
[°C]		[°C]		[°C]		[°C]	

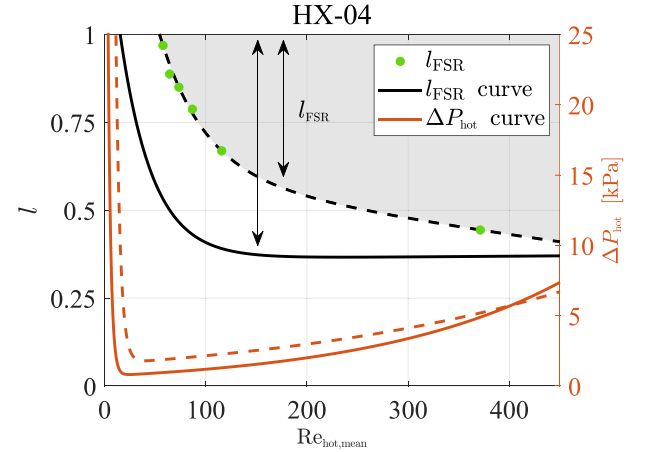


Fig. 11. FSR and ΔP_{hot} curves for HX-04 ($D_{\text{cold}} : 2 \text{ mm}$, $D_{\text{hot}} : 3 \text{ mm}$) as a function of $Re_{\text{hot,mean}}$ (dashed line: $\dot{m}_{\text{cold}} = 0.2 \text{ kg/min}$, solid line: $\dot{m}_{\text{cold}} = 0.1 \text{ kg/min}$).

The size of FSR under $Re_{\text{hot,mean}} = 100$ for each PCHE with different mass flow rate of cold channel are listed in Table 11. HX-02 had the largest FSR area of the four PCHEs. As the mass flow rate of cold channel decrease by 50 % ($\dot{m}_{\text{cold}} = 0.2 \text{ kg/min} \rightarrow 0.1 \text{ kg/min}$), size of FSR increase by approximately 2.1 to 7.3 times. Similarly, increasing the diameter of the hot channel by 1.5 times ($D_{\text{hot}} : 2 \text{ mm} \rightarrow 3 \text{ mm}$) results in FSR increase of 1.01 to 3.3 times.

Since FSR provide a criterion for avoiding freezing phenomenon inside PCHEs, further analyses are required to reflect the grown of solid layer. Bi/Bi_{freeze} should be greater than 1 to satisfy the condition for commencement of solidification. However, as shown in the section 4.2 and section 4.3, the contour map of the Bi/Bi_{freeze} does not show any areas with values greater than 1. This can be explained by the following reasons: 1) A modified Nu correlation is required for heat exchangers with multiple channels under cryogenic conditions. The heat transfer coefficients used to calculate the Bi/Bi_{freeze} were obtained from Eq. (3) known as the Dittus Boelter correlation. The literature [41] suggests that for a multi-channel heat exchanger under cryogenic conditions, the Nu

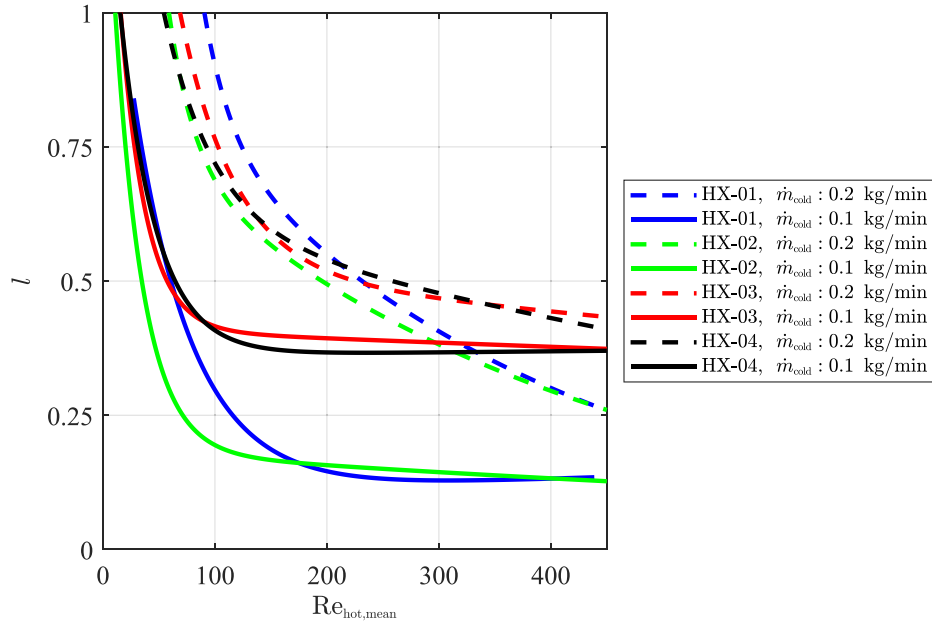


Fig. 12. FSR curves for each four PCHE as a function of $Re_{hot,mean}$ (dashed line: $\dot{m}_{cold} = 0.2$ kg/min, solid line: $\dot{m}_{cold} = 0.1$ kg/min).

Table 11

FSR variation of different PCHEs under $Re_{hot,mean} = 100$.

	HX-01 ($D_{cold} : 1$ mm, $D_{hot} : 2$ mm)		HX-02 ($D_{cold} : 1$ mm, $D_{hot} : 3$ mm)	
	$\dot{m}_{cold} = 0.2$ kg/min	$\dot{m}_{cold} = 0.1$ kg/min	$\dot{m}_{cold} = 0.2$ kg/min	$\dot{m}_{cold} = 0.1$ kg/min
l_{FSR}	0.096	0.701	0.313	0.805
	HX-03 ($D_{cold} : 2$ mm, $D_{hot} : 2$ mm)		HX-04 ($D_{cold} : 2$ mm, $D_{hot} : 3$ mm)	
	$\dot{m}_{cold} = 0.2$ kg/min	$\dot{m}_{cold} = 0.1$ kg/min	$\dot{m}_{cold} = 0.2$ kg/min	$\dot{m}_{cold} = 0.1$ kg/min
l_{FSR}	0.236	0.585	0.282	0.592

value may be greater than the value calculated using the Dittus-Boelter correlation. Therefore, the modified Nu correlation should be used. 2) Some literatures [34,42] suggest that the Nu correlation for the freezing phenomenon should reflect the influence of the solid layer thickness. The modified Nu correlation is expressed as follows:

$$Nu = 0.0155Re^{0.83}Pr^{0.5}\left(\frac{1}{r_f^{0.83}}\right) \quad (19)$$

When the solid layer grows, the radius of the liquid region, r_f , decreases (Fig. 4). As a result, the Nu obtained from Eq. (19) increases due to the contribution of both the Re term and r_f term. However, the internal sections of PCHE are not visible, so a new method is needed to account for the influence of the solid layer. 3) In the case of a multi-channel, the flow does not occur uniformly, which causes a heat transfer difference between the channels [43]. Therefore, the effects of a non-uniform flow distribution should be considered when analyzing the freezing phenomenon inside PCHE. To find the operating conditions that avoid the freezing phenomenon, further research is needed to find the modified Nu correlation for PCHE under cryogenic conditions.

4.5. Selecting hot fluids used in cryogenic heat exchangers

This section discusses various fluids that can be used as hot fluids in heat exchangers for cryogenic applications. There are many factors to consider when selecting a hot fluid for a cryogenic heat exchanger, including the freezing phenomenon, thermal conductivity, specific heat, viscosity, pumping power, fluid cost, manufacturing heat exchanger

cost, chemical hazards, etc. The physical properties of fluids used in cryogenic heat exchangers are listed in the Table 12.

One of the most common applications of PCHEs under cryogenic conditions is the liquefied natural gas (LNG) regasification process. Water and EGW are commonly used as the hot fluid [45]. The advantage of using water as a hot fluid is its high thermal conductivity and heat capacity compared to other fluids. However, water has a high freezing point which is critical to avoid the freezing phenomenon [46]. EGW and PGW, which have lower freezing points, could be an alternative. As the concentration of the mixture increases, it has a lower freezing point and at the same concentration, EGW has a lower freezing point than PGW, making it more effective for freeze avoidance [47]. In terms of cost, EGW is also less expensive than PGW [48]. However, EGW has a lower heat capacity than PGW and is more toxic to humans [49]. PGW has the advantage of being chemically safe [47] and having a high heat capacity, but it is expensive and has a high viscosity. Viscosity is a main parameter in determining pumping power (PP) [50]. The pumping power is the product of the differential pressure and the volumetric flow rate expressed as Eq. (20).

Table 12

Thermophysical properties of water, EGW, propylene-glycol water (PGW) and silicone fluid at 20 °C.

Working fluid [wt. %]	Density [kg/m ³]	Viscosity [μPa·s]	Thermal conductivity [W/m·K]	Heat capacity [kJ/kg·K]	Freezing Temp [°C]
Water	998	1002	0.598	4184	0
EGW 40 %	1052	2819	0.425	3519	−24
EGW 50 %	1065	3693	0.389	3312	−36
EGW 60 %	1077	4845	0.357	3098	−51
PGW 40 %	1032	4384	0.400	3707	−21
PGW 50 %	1039	6394	0.359	3530	−32
PGW 60 %	1043	9279	0.322	3340	−50
Silicone Fluid [44]*	877	2138	0.114	1701	−80

* Neo Silicone Fluid M−2.

$$PP_{\text{hot}} = \frac{\Delta P_{\text{hot}} \dot{m}_{\text{hot}}}{\rho_{\text{hot}}} = f \text{Re}_{\text{hot}} \frac{\mu_{\text{hot}} L}{2A_{\text{cross,hot}}} \left(\frac{\dot{m}_{\text{hot}}}{\rho_{\text{hot}} D_h} \right)^2 \quad (20)$$

$A_{\text{cross,hot}}$ in the right term of the Eq. (20) is the cross-sectional area of the flow channel. Because of the larger viscosity compared to EGW, PGW requires more pumping power. Another option is silicone fluid. The silicone fluid is widely used for hydrogen pre-cooling at gaseous hydrogen refueling stations [51]. Compared to EGW, silicone fluid has a very low freezing point, which is advantageous for the freezing phenomenon [52]. However, it has the disadvantage of low thermal conductivity and heat capacity which means that the size of the heat exchanger should be increased compared to the hot fluids mentioned before, and the cost of manufacturing the heat exchanger will also increase. In addition, these fluids have temperature-dependent properties that must be considered. One of the characteristics of cryogenic heat exchangers is the rapid temperature change of the hot fluid. Fig. 13 shows the change in viscosity and thermal conductivity with the temperature for each fluid. For example, EGW 50 % has a viscosity of 3693 $\mu\text{Pa}\cdot\text{s}$ at 20°C and 22178.4 $\mu\text{Pa}\cdot\text{s}$ at -20°C which is 6 times higher. Moreover, as the operating temperature changes, superiority of the thermo-physical properties is changed. For instance, the viscosity of EGW 60 % is higher than that of PGW 40 % when temperature exceeds 10°C, but this trend reverses when temperature drops below 10°C. These results emphasize the importance of considering the temperature-dependent changes in fluid properties when choosing the hot fluid for cryogenic PCHE.

As described in this section, selecting a hot fluid of PCHE for cryogenic applications requires a complex analysis that not only compares the physical properties of the fluids, but also takes into account the design difficulty and manufacturing cost due to the increased size of the heat exchanger, the operating temperature conditions, and more. Further research is needed to select the hot fluid of PCHEs for cryogenic applications.

5. Conclusions

This work presents an experimental and theoretical study on the freezing phenomenon in PCHE for cryogenic applications. The Bi number and the FSR were introduced to help understand the freezing phenomenon.

- Experiments were performed to observe the freezing phenomenon inside PCHEs. Four types of lab-scale PCHEs with semi-circular channels of different diameters were fabricated. Nitrogen was used as the cold fluid, while EGW (50 wt%) was used as the hot fluid. Experiments were conducted by reducing the flow rate of the hot fluid while maintaining the flow rate of the cold fluid. It was found that when the flow rate of the hot fluid was reduced to a certain value, the freezing phenomenon occurred. The differential pressure in the hot channel increased rapidly due to the formation of a solid layer inside the channel.
- Based on the experimental data, the freezing phenomenon was analyzed in terms of a dimensionless number. The growth rate of the solid layer can be expressed as a function of the Bi number, and $\text{Bi}_{\text{freeze}}$ was defined as the point at which the growth rate of the solid layer becomes zero. The $\text{Bi}/\text{Bi}_{\text{freeze}}$ value indicates whether the freezing phenomenon occurs. Under various experimental conditions, contour maps of the $\text{Bi}/\text{Bi}_{\text{freeze}}$ for each PCHE were obtained. The Freeze Safety Region (FSR), the region where the freezing phenomenon does not occur, increases in size as the flow rate of the cold fluid decreases, and the cross-sectional area of the hot channel increases. Therefore, it is recommended to choose operating conditions that maximize the size of the freeze safety region, in order to avoid the freezing phenomenon in PCHEs.

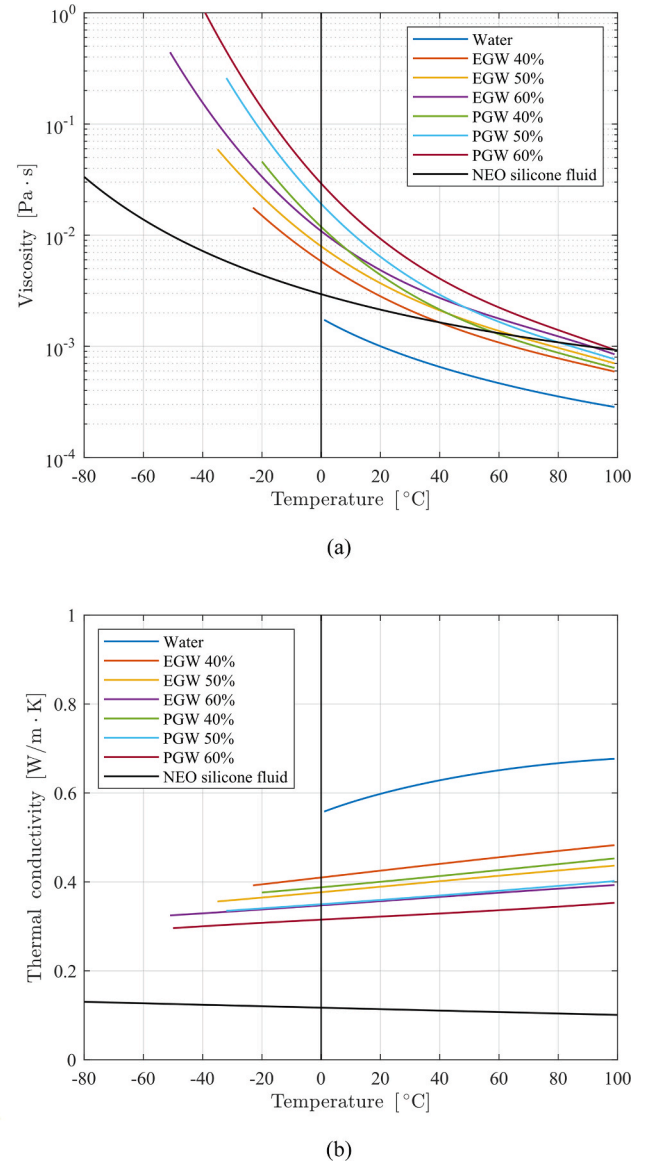


Fig. 13. Fluid properties-temperature changes for the EGW, PGW and NEO silicone fluid (a) Viscosity-temperature curves (b) Thermal conductivity-temperature curves.

- FSR enables a quantitative comparison of the ability to avoid freezing phenomenon inside PFHEs. For example, in HX-04 ($D_{\text{cold}} = 2$ mm, $D_{\text{hot}} = 3$ mm) with $\text{Re}_{\text{hot, mean}}$ of 100, when $\dot{m}_{\text{cold}}$ decreases 0.2 to 0.1 kg/min, the FSR increases by 1.8 times. In other case, comparing HX-01 ($D_{\text{cold}} = 1$ mm, $D_{\text{hot}} = 2$ mm) and HX-02 ($D_{\text{cold}} = 1$ mm, $D_{\text{hot}} = 3$ mm), when D_{hot} increases by 1.5 times, the FSR increases by 1.16 times. As a result, when the mass flow rate of cold channel decrease 0.2 kg/min to 0.1 kg/min, size of the FSR increase by 2.1 to 7.3 times. Similarly, As the diameter of the hot channel increases from 2 mm to 3 mm, the size of the FSR increases by 1.01 to 3.3 times.
- When selecting the hot fluid for PCHE in cryogenic applications, PGW, silicone fluid, and other fluids can be used as alternatives to EGW. To avoid the freezing phenomenon, a fluid with a lower freezing point is preferable. In addition to avoiding freezing phenomenon, many other factors must be considered, including thermal conductivity, specific heat, viscosity, pump power, manufacturing costs, etc.

CRediT authorship contribution statement

Bogyom Kim: Conceptualization, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing. **Kong Hoon Lee:** Supervision, Project administration. **Min-chang Kim:** Validation, Formal analysis, Investigation. **Sunyoung Moon:** Formal analysis, Investigation. **Jin Woo Yoo:** Formal analysis, Investigation. **Woogyong Kim:** Conceptualization, Validation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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